

## CONTACT

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- 📍 Turkey, Gaziantep
- 🌐 <https://louai-aldalati.github.io/louai>

## EDUCATION

2023 – 2027

MANISA CELAL BAYAR UNIVERSITY

- Bachelor's in Software Engineering | GPA: 3.68 / 4.00

2020 – 2023

ÇAMLIK ANADOLU LİSESİ

- High School – Science Stream

## SKILLS

- **Programming Languages:** Python, Java, C++, C, JavaScript, SQL.
- **Web & Databases:** Django, Django REST Framework, API Development, Relational Databases, Stored Procedures & Triggers.
- **AI & Machine Learning:** NLP, Computer Vision, RAG, Fine-Tuning, Transfer Learning.
- **Models & Tools:** Llama 3, MobileNetV2, XGBoost, Git, GitHub, Agile/Scrum.
- **Soft Skills:** Analytical Thinking, Teamwork, Effective Communication, Continuous Learning.

## LANGUAGES

- Arabic - Native
- English - Advanced
- Turkish - Fluent

## REFERENCES

- Assoc. Prof. Dr. Fatih Yücalar  
Head of Computer Software  
Department Hasan Ferdi Turgutlu  
Faculty of Technology, Manisa Celal  
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Email: [fatih.yucalar@cbu.edu.tr](mailto:fatih.yucalar@cbu.edu.tr)
- Asst. Prof. Dr. İrfan Aygün  
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# LOUAI ALDALATI

## SOFTWARE DEVELOPER

## PROFILE SUMMARY

Software Engineering student specializing in developing integrated software solutions that merge the power of Backend (Django) with AI models (LLMs) and Computer Vision. Characterized by analytical thinking and fluency in three languages (Arabic, English, Turkish). Highly motivated to leverage modern technologies to innovate efficient systems and solve complex real-world problems.

## TECHNICAL PROJECTS

### AI Hotel Chatbot

2026

Technologies: Llama 3 (8B Instruct), RAG, Fine-Tuning, NLP

- Developed an advanced intelligent assistant based on the open-source Llama 3 model to provide accurate and automated responses.
- Applied Retrieval-Augmented Generation (RAG) and Fine-Tuning to fully customize model responses based on specific hotel policies and data, enhancing customer service efficiency.
- Source Code : [https://github.com/louai-aldalati/sunrise\\_hotel](https://github.com/louai-aldalati/sunrise_hotel)

### Real-Time Emotion Recognition

2026

Technologies: Computer Vision, MobileNetV2, Transfer Learning, Python

- Built a local computer vision system that analyzes live webcam feeds to classify 8 different human emotions with high accuracy.
- Utilized Transfer Learning with the MobileNetV2 architecture, fine-tuning final layers to accelerate performance and reduce resource consumption.
- Source Code : <https://github.com/louai-aldalati/Emotion-Recognition-MobileNetV2>

### Intelligent Cloud HR (Candidate Filter)

2026

Technologies: Python, Django, spaCy (NLP), Tesseract OCR, PostgreSQL, Docker, Railway PaaS

- Developed a cloud-based HR platform leveraging Natural Language Processing (NLP) and Optical Character Recognition (OCR) to automate CV data extraction and generate an objective "Match Score" against job criteria.
- Built the backend using Django and PostgreSQL, and deployed the system via Docker containers on the Railway PaaS to ensure security and scalability.
- Source Code: [https://github.com/louai-aldalati/candidate\\_filter](https://github.com/louai-aldalati/candidate_filter)

### Smart Resource Allocation System

2026

Technologies: Deep Reinforcement Learning (DQN), Python, PyTorch, Gymnasium, Algorithm Design

- Engineered a simulated cloud environment to solve the dynamic 2D Knapsack problem, optimizing CPU and RAM allocation for incoming tasks.
- Trained a Deep Q-Network (DQN) agent with comprehensive hyperparameter optimization, outperforming traditional heuristic (Greedy) algorithms and achieving 90.3% of the theoretical maximum (Dynamic Programming).
- Source Code: <https://github.com/selinkincal/smart-resource-allocation-rl>

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Email: [irfan.aygun@cbu.edu.tr](mailto:irfan.aygun@cbu.edu.tr)

## TECHNICAL PROJECTS

### Weather Forecasting ML API

2026

Technologies: Machine Learning, XGBoost, Time-Series Analysis, Django, DRF

- Designed and trained a machine learning model using XGBoost to analyze time-series data (temperature, humidity, wind direction) for future value prediction.
- Developed the backend using Django, providing the AI model as a service via Django REST API for seamless integration with web or mobile apps.
- Source Code : <https://github.com/louai-aldalati/weather-predictor>

### Pawly: Pet Care Marketplace Mobile App

2026

Technologies: Flutter, Dart, Firebase (Auth, Firestore, FCM), Google Maps API, NoSQL

- Developed a comprehensive cross-platform mobile marketplace connecting pet owners with service providers, featuring a dynamic multi-role architecture (Pet Owner, Service Provider, Hybrid) within a single application.
- Engineered a real-time backend utilizing Firebase Firestore, and seamlessly integrated advanced features including Google Maps API with Geocoding for location tracking, and Firebase Cloud Messaging (FCM) for instant push notifications.
- Source Code: [https://github.com/selinkincal/pet\\_care\\_app](https://github.com/selinkincal/pet_care_app)

### Kelime6: Language Learning Platform

2025

Technologies: Django, Python, Relational Database, Git, Scrum

- Programmed an educational system based on the Spaced Repetition System (SRS) algorithm to schedule word appearances based on user performance.
- Developed modules for user authentication, analytical dashboards, and a dynamic testing system including an interactive Wordle game.
- Source Code : <https://github.com/louai-aldalati/kelime6>

### TinyHouse: Rental Management System

2025

Technologies: Django, SQL (Stored Procedures, Triggers), RBAC

- Developed a property management platform supporting advanced Role-Based Access Control (RBAC) for Admins, Hosts, and Guests.
- Engineered a normalized relational database utilizing Stored Procedures and Triggers to automate booking and payment updates.
- Source Code : <https://github.com/louai-aldalati/tinyhouse>

## TRAINING COURSES

### Monitoring and Evaluation in NGOs /trainer Khalil M. Hasno

2023

- Covers key concepts in performance monitoring and evaluation (M&E) and provides practical insights into professional reporting processes

### Developing Entrepreneurship and Innovation / INJAZ Lebanon

2019

- Addresses the fundamental traits of entrepreneurship and critical success factors for establishing and managing new projects

### Life Skills / AVSI

2019

- Explores core topics regarding human nature, psychological structure, and personal development

## HOBBIES

• **Chess:** Enhancing strategic thinking and problem-solving skills

• **Programming:** Constantly building side projects and exploring new frameworks

• **Cycling:** Maintaining physical fitness and endurance

• **Sports:** General interest in fitness and athletic activities